

ASSESSMENT MARKING GUIDE

QUESTION 1

- **(a) [Achieved]:** Shows correct substitutions: $10.0 = 0 + \alpha(4.00) \Rightarrow \alpha = 10.0/4.00 = 2.50 \text{ rad s}^{-2}$.
- **(b) [Merit]:** Shows clear two-step calculation:
Net torque: $\tau_{\text{net}} = I\alpha = 1.50 \times 2.50 = 3.75 \text{ N m}$
Applied torque: $\tau_{\text{applied}} = \tau_{\text{net}} + \tau_{\text{friction}} = 3.75 + 0.50 = 4.25 \text{ N m}$. Allow correct unit N m or $\text{kg m}^2 \text{ s}^{-2}$.
- **(c) [Excellence]:** Identifies **Conservation of Angular Momentum**. Explains that because the rotational inertia (I) increases, angular velocity (ω) must decrease to keep total angular momentum ($L = I\omega$) constant.

QUESTION 2

- **(a) [Achieved]:** Identifies **Gravity (Weight)** and **Tension** (Do not accept centripetal).
- **(b) [Merit]:** Correctly resolves components to find F_c :
Vertical balance: $F_T \cos \theta = F_g \Rightarrow F_T \cos(52.0^\circ) = mg = 0.150 \times 9.80 = 1.47 \text{ N} \Rightarrow F_T = 2.39 \text{ N}$
Horizontal force: $F_c = F_T \sin(52.0^\circ) = 2.39 \times \sin(52.0^\circ) = 1.88 \text{ N}$.
Alternative shortcut: $F_c = mg \tan \theta = 1.47 \times \tan(52.0^\circ) = 1.47 \times 1.280 = 1.88 \text{ N}$.
- **(c) [Excellence]:** Shows clear speed calculation using F_c :
Formula step: $F_c = \frac{mv^2}{r} \Rightarrow v = \sqrt{\frac{F_c r}{m}} = \sqrt{\frac{1.88 \times 0.500}{0.150}} = \sqrt{6.27} \approx 2.50 \text{ m s}^{-1}$.
Explains qualitative change: If speed (v) increases, the required centripetal force ($F_c = \frac{mv^2}{r}$) increases. Because the horizontal component of the tension force ($F_T \sin \theta = mg \tan \theta$) provides this centripetal force, the angle θ must increase to make $\tan \theta$ larger.

QUESTION 3

- **(a) [Achieved]:** Identifies **Gravity / Gravitational Force**.
- **(b) [Merit]:** Calculation using M_E : $v = \sqrt{\frac{GM_E}{r}} = \sqrt{\frac{6.67 \times 10^{-11} \times 5.97 \times 10^{24}}{7.00 \times 10^6}} = \sqrt{5.69 \times 10^7} \approx 7540 \text{ m s}^{-1}$.
- **(c) [Excellence]:** Explains that the center of any orbital plane must align with Earth's center of mass. An orbit centered over Auckland would require continuous lateral thrust because gravity pulls directly toward Earth's center.